

Tushar Sharma

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EDUCATION

Bachelor of Science (BS) in Data Science and Applications

2022 – present | Chennai, Tamil Nadu

Indian Institute of Technology, Madras

Relevant Courses - Mathematics & Statistics for Data Science, Business Data Management, Python Programming, Business Analytics, Database Management System (DBMS), Machine Learning

TECHNICAL SKILLS

Languages & Tools

Python, Relational Database (MySQL & PostgreSQL), HTML, CSS, Conda, Version Control System (Git & DVC), Microsoft Office, Amazon Sagemaker Studio Lab, JupyterLab, Docker, DagsHub, VS Code

Libraries & Frameworks

Pandas, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, Keras, PyTorch, SpaCy, NLTK, Streamlit, Flask, BeautifulSoup, LangChain, Transformers (Hugging Face), FastAPI, MLflow, Optuna

AI & Data Science Stack

Data Acquisition-Manipulation-Visualization, Supervised-Unsupervised Learning, Deep Learning, MLOps, Feature Engineering, Natural Language Processing (NLP), Time-Series Forecasting, Hyperparameter Optimization, Transfer Learning, Large Language Model (LLM)

PROFESSIONAL EXPERIENCE

Freelance ML Application Developer

May 2023 – Jul 2023 | Chennai (Remote)

Cisco ThingQbator | Nasscom Foundation

- Developed a high-accuracy (95%) proof-of-concept (POC) for a plant disease detection web application using quantized CNN models for an AgTech startup. Optimized model for minimal resource consumption and deployed on Azure (Standard_DS3_v2) VM instance.
- Designed a responsive user interface (HTML/CSS/JavaScript) with Flask backend for seamless user interaction across devices. [Project Repository - tushar-mahalya/Krishi](#)

Data Analyst - Intern

Nov 2022 – Jun 2023 | Kanpur (Hybrid)

QualDigiIn Technologies® Pvt. Ltd.

- Cleansed and organized client data using MS Excel to ensure data integrity for advanced analysis.
- Maintained and streamlined SQL queries for multiple clients, facilitating smooth integration with LAMP stack business websites.

Data Science & Analytics - Virtual Intern

Jul 2022 – Sep 2022 | Remote

Forage® Pvt. Ltd.

- Utilized a diverse toolkit (PowerPoint, Python, SQL, Tableau, Time Series Analysis, Machine Learning, Quantization) to perform in-depth data analysis for multiple short-term Virtual Experience and internship projects for esteemed organizations (Quantum, Accenture, JP Morgan, BCG, British Airways & Cognizant).
- Delivered impactful results through detailed reports, data visualizations, and strategic plans, leading to measurable improvements in engagement metrics, conversion rates, cost efficiency, and supply chain optimization.

COURSES & CERTIFICATIONS

Machine Learning (Andrew NG)

Aug, 2022 - Oct, 2022

Stanford University | Coursera

AI-2 : Convolutional NN

Jun, 2022 - Jul, 2022

Univ.AI

IBM Data Science

Oct, 2022 - Dec, 2022

IBM | Coursera

Google Data Analytics

May, 2021 - Sep, 2021

Google | Coursera

PROJECTS

Spotify Song Recommender System [↗](#)

github.com/tushar-mahalya/Songs-Recommender-System

Aim: To develop a web application that emulates the UI of Spotify, integrating a robust recommender system that utilize the audio features of selected songs to provide personalized recommendations and Analytical Engine capable of analyzing songs & genres.

- Scraped the list of songs in Billboards Hot 100 Charts spanning from 1946 to 2022 (77 years) using BeautifulSoup and fetched metadata, including audio features, for each song using the Spotify Web API, creating dataset with over 6500+ songs.
- Preprocessed relevant features using TF-IDF, TextBlob, and OHE techniques to generate a Song Summarization Vector, and employed Cosine Similarity to recommend songs from the selected songs list.
- Web Application was successfully deployed on Streamlit Cloud, ensuring easy access and smooth usage for all users.

Custom ChatGPT [↗](#)

github.com/tushar-mahalya/Custom-ChatGPT

Aim: Develop a customized chatbot integrated with the GPT-3 model, trained and fine-tuned on a custom text corpus. This tailored chatbot will generate contextually appropriate, human-like responses closely aligned with the training data.

- Collected comments from the top 1000 posts of the three leading data science communities on Reddit using the official Reddit API. This resulted in a text corpus of approximately 12 million words from around 223k comments.
- Created indexes of GPT-3 model embeddings using the LangChain and FAISS framework to achieved contextually appropriate responses to user queries/prompts.
- Developed a simulation web application using Streamlit for a user-friendly chatbot experience.

VOLUNTEERSHIP

Academic Societies

Jul 2022 – present

Volunteer

Active participant in the Apostrophe Oratory Club, Corbett House #206 & Ramanunjan Society for Research at IIT-M, actively participating and moderated in group activities and fostering a sense of community.

Muskurahat Foundation

Nov 2022 – Jan 2023

Fundraising Intern

Assisted fundraising initiatives in a team for underprivileged children's education through crowd-funding under Project 'KEYtaab', acting as an affiliate and designing social media campaigns.

Dayanand Anglo-Vedic (DAV) College

Apr 2022 – May 2022

Accessibility Scribe

Provided exam-time support to multiple Z visually impaired and physically challenged students, ensuring their smooth participation and promoting inclusivity.

ACHIEVEMENTS

- Ranked 1st (out of 950 participants) with an accuracy of 80.4% in the Machine Learning Capstone Project competition at IIT-M. This project involved classifying recipe review ratings from multiple users. Employed rigorous feature engineering, utilizing a Stacking Ensemble of HP optimized models.
- Achieved a top 2% (33rd place out of 2,199 teams) finish on the Kaggle Competition - Steel Plate Defects Prediction in a team of two. Developed a Mixed Meta-Classifer Ensemble of HP optimized models to predict the probability of defects in steel plates, resulting in an AUC-ROC score of 0.889.
- Ranked 40th out of 600+ Data Scientists in the TechnoPro Design Food Package Image Analysis Challenge (SIGNATE.jp). Developed a SWIN Transformer model (PyTorch) for automated food/drink package classification, addressing logistics/inventory optimization. Achieved 0.837 AUC-ROC.
- Received the "Academic Excellence" award from O.P. Verma, District and Session Judge, Kanpur, for achieving a perfect 10 CGPA in high school.